

ST PETER-ORDING, Germany

WMO: 100280

Lat: 54.3278N

Lon: 8.6031E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.3	-5.2	-9.8	1.6	-5.9	-8.1	1.9	-4.5	19.6	5.0	17.9	5.9	4.1	100	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	26.7	19.4	24.3	18.4	22.4	17.7	20.2	24.6	19.3	23.0	18.4	21.6	5.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.5	22.0	17.7	12.7	20.9	17.1	12.2	20.3	57.8	24.8	54.9	23.2	52.1	21.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.3	12.8	-9.4	30.3	3.2	1.5	-11.7	31.4	-13.6	32.2	-15.4	33.0	-17.7	34.1		
(5)			-9.7	22.2	3.3	1.2	-12.0	23.0	-13.9	23.8	-15.8	24.4	-18.2	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	1.8	1.5	4.0	8.5	12.2	15.2	17.9	17.9	15.1	10.3	6.6	2.7		
	DBStd	6.64	3.75	3.36	3.26	3.09	2.65	2.60	2.39	2.21	2.30	3.17	3.09	3.70		
	HDD10.0	1121	256	238	186	64	9	0	0	0	1	36	106	227		
	HDD18.3	3273	514	471	444	294	192	99	39	35	99	248	352	485		
	CDD10.0	944	0	0	1	21	76	157	244	244	153	45	4	0		
	CDD18.3	54	0	0	0	0	1	7	24	21	2	0	0	0		
	CDH23.3	297	0	0	0	1	6	48	138	97	7	0	0	0		
	CDH26.7	54	0	0	0	0	0	4	30	19	0	0	0	0		
(14) Wind	WSAvg	6.4	7.2	6.5	6.5	5.8	6.2	6.5	6.1	6.0	6.4	6.4	6.6	6.5		
(15) Precipitation	PrecAvg	792	65	47	43	36	46	63	82	94	84	86	73	73		
	PrecMax	1024	132	110	83	105	92	117	147	174	174	182	176	161		
	PrecMin	494	2	16	7	0	6	1	26	28	21	26	4	16		
	PrecStd	116	35	21	23	21	20	31	36	42	43	38	39	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.1	14.7	21.8	23.9	27.5	29.8	29.0	24.2	18.2	13.7	10.2		
		MCWB	8.5	7.6	9.4	13.3	16.4	18.8	20.3	19.8	18.1	15.6	12.3	9.0		
	2%	DB	8.3	7.8	11.4	18.7	21.4	24.7	27.0	26.1	22.0	16.6	12.2	8.9		
		MCWB	7.2	6.9	8.1	12.0	15.2	18.1	19.8	19.4	17.3	14.5	11.0	7.9		
	5%	DB	7.4	6.8	9.0	16.3	18.9	21.6	24.4	23.4	20.0	15.6	11.4	7.9		
		MCWB	6.5	5.9	7.1	11.4	13.6	16.9	18.9	18.6	16.5	13.7	10.2	7.0		
	10%	DB	6.6	5.9	7.7	13.7	16.9	19.5	22.4	21.5	18.5	14.6	10.5	7.2		
		MCWB	5.9	5.1	6.5	10.3	12.6	15.8	18.3	18.0	15.9	13.0	9.4	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	8.1	10.2	14.7	18.0	20.5	21.5	21.3	19.2	16.1	12.6	9.2		
		MCDB	9.4	8.6	12.7	20.4	23.2	25.5	27.1	26.0	22.5	18.1	13.4	9.9		
	2%	WB	7.5	7.0	8.6	12.8	15.6	18.8	20.4	19.8	17.8	14.9	11.2	8.0		
		MCDB	8.1	7.7	10.7	16.8	19.3	22.8	25.0	23.7	20.7	16.1	12.0	8.6		
	5%	WB	6.7	6.1	7.5	11.7	14.5	17.3	19.5	19.0	17.0	14.1	10.4	7.2		
		MCDB	7.2	6.6	8.9	15.4	17.7	20.6	23.5	22.4	19.3	15.2	11.2	7.7		
	10%	WB	6.0	5.2	6.7	10.5	13.4	16.1	18.5	18.3	16.3	13.2	9.6	6.5		
		MCDB	6.5	5.7	7.6	13.5	16.2	19.0	21.8	21.1	18.3	14.4	10.4	7.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	3.4	4.8	7.0	6.4	5.8	5.8	5.6	5.3	4.7	3.3	3.5		
		MCDBR	2.5	3.5	7.6	12.2	11.2	10.7	11.3	10.3	8.3	5.3	3.0	2.6		
	5% WB	MCWBR	2.6	3.0	5.1	6.8	6.4	5.8	5.4	4.9	4.8	3.8	2.8	2.7		
		MCWBR	2.6	2.9	6.5	10.3	9.1	9.2	9.5	8.0	6.6	4.7	2.9	2.7		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.346	0.372	0.405	0.397	0.395	0.411	0.408	0.387	0.364	0.341	0.319		
		taud	2.291	2.320	2.284	2.248	2.334	2.362	2.332	2.348	2.392	2.397	2.336	2.288		
	Ebn at Noon		607	733	796	819	850	854	830	809	777	705	588	534		
		Edh at Noon	59	80	102	121	117	115	116	108	92	74	57	49		
(44) All-Sky Solar Radiation	RadAvg		0.52	1.17	2.44	4.07	5.15	5.34	5.04	4.11	2.94	1.59	0.65	0.38		
	RadStd		0.06	0.17	0.22	0.46	0.52	0.44	0.61	0.32	0.33	0.18	0.08	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-131	N/A	N/A

Nomenclature: See separate page